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A Habituation Sensory Nervous System with Memristors

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The sensory nervous system (SNS) builds up the association between external stimuli and the response of organisms. In this system, habituation is a fundamental characteristic that filters out irrelevantly repetitive information and makes the SNS adapt to the external environment. To emulate this critical process in electronic devices, a Li_xSiO_y-based memristor ($TiN/Li_xSiO_y/Pt$) is developed where the temporal response under repetitive stimulation is similar to that of habituation. By connecting this synaptic device to a leaky integrate-and-fire neuron based on a Ag/SiO_2:Ag/Au memristor, a fully memristive SNS with habituation is experimentally demonstrated. Finally, a habituation spiking neural network based on the SNS is built and its application in obstacle avoidance for robot navigation is successfully presented. The results provide that a direct emulation of the biologically inspired learning process by memristors could be a sound choice for neuromorphic hardware implementation.

The sensory nervous system (SNS) is the cornerstone of the human biological system. It builds up the connection between the external environment and organisms. The external information is first perceived by various receptors (haptic, optical, olfactory, etc.), then conveyed to the brain and processed. Finally, corresponding responses to the external information are performed.[1] However, in the environment, various kinds of stimuli are received by the SNS at all times. Therefore, it is of great importance that organisms are able to distinguish stimuli that require attention and those that are irrelevant and should not be responded to. Even simple creatures must continually discriminate stimuli that are worthy of a response and those that have no behavioral relevance.[2]

In the SNS, habituation is one of the fundamental forms of such discrimination. A brief definition of habituation is the decreased response to identically repeated stimuli, though it may also include other complex characteristics.[3,4] Habituation plays an imperative role for the organism to adapt to the general environment and helps organisms to filter out irrelevantly repetitive information.[5,6] Therefore, incorporating habituation into artificial SNS based on electronic devices would significantly improve its information processing capability.

Synapses and spiking neurons are the fundamental units of artificial SNS. Generally, complementary metal–oxide–semiconductor (CMOS) circuits are developed to mimic the functions of synapses and neurons.[7,8] However, because CMOS devices lack intrinsic biological resemblance, the constructed circuits are always complex, which limits the power and area efficiencies.[9] In recent years, emerging devices, e.g., resistive random-access memory,[10–14] phase change memory,[15–19] ferroelectric random-access memory,[20–23] van der Waals heterostructures,[24,25] and synaptic transistors,[26–28] have been successfully demonstrated for the emulation of biological synapses and neurons. The advantage of the emerging devices is that the synaptic or neural functions can be mimicked at least partially by a single device, greatly reducing the circuit complexity. Hence, developing artificial SNS with emerging devices has drawn significant attention in the academic community.[29–31] Recently, artificial sensory afferent nerve based on flexible transistors and emerging devices have been studied for converting analog sensory signals into spikes similar as that in biological systems.[29,32] The sensorimotor synapse was also investigated to construct motor nervous system.[33] The realization of these unit paves the way for constructing artificial SNSs. Furthermore, haptic memory array, which enables organisms to figure out how much force needs to hold objects, was achieved by connecting the memristor array with pressure sensor.[34] Visual memory array, which is an essential exteroceptive sensory memory for cognitive tasks, was achieved by connecting the memristor array with UV light sensor.[35,36] Although these works may represent significant steps toward the realization of bio-inspired SNS, habituation has not been incorporated into the SNS yet, nor demonstrated its application in systems.

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In this work, a fully memristor-based SNS that consists of memristor-based synapse (TiN/Li_xSiO_y/Pt) and Ag-based neuron is proposed. Habituation is successfully realized in the Li_xSiO_y-based synaptic device, in addition to the conventional synaptic plasticity, such as long-term potentiation/depression and spike-rate dependent plasticity. The origin of synaptic behaviors may be attributed to the formation and dissolution of the channel with lithium titanium silicate ($LiTiSiO_4$) phase, as confirmed by the high-resolution transmission electron microscopy (HRTEM). The formation and rupture dynamics of the $LiTiSiO_4$ channel is analogous to the activation and inactivation of the Ca^{2+} channel in biological SNS, leading to habituation.[37] Then, the synaptic device is connected to a leaky integrate-and-fire (LIF) neuron and a simple SNS with habituation characteristics is constructed. To further demonstrate the application of the SNS, a habituation spiking neural network (HSNN) with a habituation learning rule for obstacle avoidance in robot navigation is built. The simulation result shows that the robot with HSNN presents a significant reduction (up to 37.2%) of the absolute distance traveled around a corner. In other words, by incorporating habituation in SNS, the robot is able to filter out repetitive stimuli that might slow it down in its path course.

The biological SNS realizes information transmission and processing by conveying the outside information via sensory neurons to relay neurons, as shown in **Figure 1**a. The sensory neuron simply acts as a transducer. It transforms exteroception, e.g., sight or sound, perceived by receptor into electrical spikes. The relay neuron acts as a computing unit, which receives and processes the electrical information. The processed information is then passed to the motor neuron (not shown in the figure) to perform corresponding actions. Organisms can filter out repetitive and redundant information in the above process, benefiting from habituation mechanism. Figure 1b shows a typical response of habituation in the SNS under repeated stimuli. Biologically, habituation is typically understood by dual-process theory. In this theory, an initial sensitization process (corresponding to the response increment process) is usually presented before a habituation process (corresponding to the response decrement process).[38] The inset of Figure 1b shows the response tendency of the TiN/Li_xSiO_y/Pt device under 80 consecutive identical stimuli (3.4 V, 10 µs), which is very similar to the biological one. Since the sensory neuron in SNS is a simple transducer, we focus on emulating the computation unit, namely the relay neuron. The schematic of the fully memristor-based SNS is shown in Figure 1c. It composes of a memristive synaptic device (TiN/Li_xSiO_y/Pt) and an Ag-based LIF neuron. The fixed electrical stimulus acts as the transformed electrical information from the sensory neuron. The perceived electrical pulse signals induce the change of synaptic weight following habituation rule, then modulate the input of relay neuron (node 1 in Figure 1c), and finally result in different spiking behaviors, as will be discussed later.

Lithium silicate-based memristors exhibit uniform, fast, and analog switching behaviors, making it a promising material for neuromorphic applications.[39–41] Thus, a TiN/Li_xSiO_y/Pt device is engineered to serve as the synaptic devices in the SNS. The ingredients of the lithium silicate film are characterized by X-ray photoelectron spectroscopy (XPS). The XPS result clearly indicates the existence of Li, Si, and O elements (Figure S1a, Supporting Information) and the element ratio is Li:Si:O ≈ 2.3:1:3.4. Additionally, the crystallization of the deposited lithium silicate film is characterized by X-ray diffraction (XRD). The XRD result suggests that the film is amorphous without Li_xSiO_y and $LiTiSiO_4$ relevant peaks (Figure S1b, Supporting Information). To further confirm the amorphous state of the Li_xSiO_y film, the initial state device was characterized by

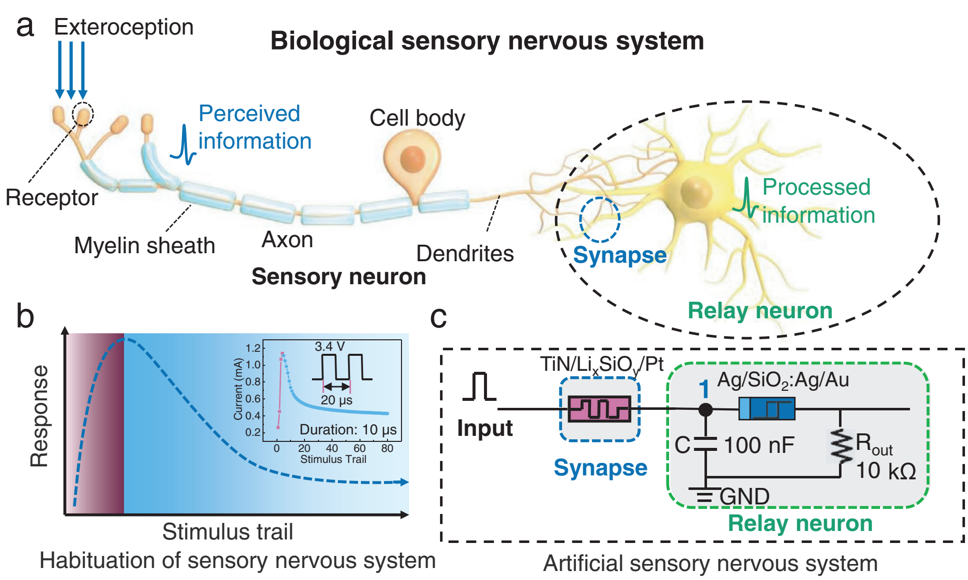

Figure 1. Schematic of the SNS. a) Biological SNS. External information is perceived by sensory neurons and then transmitted to the relay neuron for processing. b) Schematic of habitation in the SNS under the repetition of an identical stimulus. A sensitization process (pink region) usually occurs before habituation (blue region), obeying the dual-process theory. The inset shows the analogous response (current) of the TiN/Li_xSiO_y/Pt synaptic device under identical stimulus train (3.4 V, 20 µs). c) The fully memristor-based artificial SNS circuit. It consists of a synaptic device (TiN/Li_xSiO_y/Pt) and a neuron constructed by a TSM (Ag/SiO_2:Ag/Au), a capacitor (100 nF), and a resistor (10 kΩ).

HRTEM (Figure S1c, Supporting Information). The result further supports the XRD result.

The synaptic device is first characterized by DC voltage sweep. During the voltage sweep process, the Pt bottom electrode was grounded, and the TiN top electrode was biased. The device shows different resistive switching characteristics (bipolar and unipolar resistive switching) under different compliance currents (I_{cc}) (**Figure 2**). The typical bipolar resistive switching curve is obtained at the I_{cc} of 100 µA (Figure 2a). The device switches from high resistance state (HRS) to low resistance state (LRS) and retains at LRS under positive voltage sweep. A negative voltage sweep is needed to reset the device from LRS to HRS. The bipolar switching shows good retention, low cycle-to-cycle and device-to-device variations (Figure S2, Supporting Information). Whereas, when the I_{cc} is at 500 µA, the device first switches from HRS to LRS (SET), followed by a switching back process (from LRS to HRS (RESET)) under positive voltage. The negative voltage sweep for RESET is not required (Figure 2b). The SET and RESET at I_{cc} = 500 µA occurs at the same voltage polarity, which is a unipolar resistive switching behavior. In addition, the cyclability of the synaptic device under these two different I_{cc} was investigated (Figure S3,

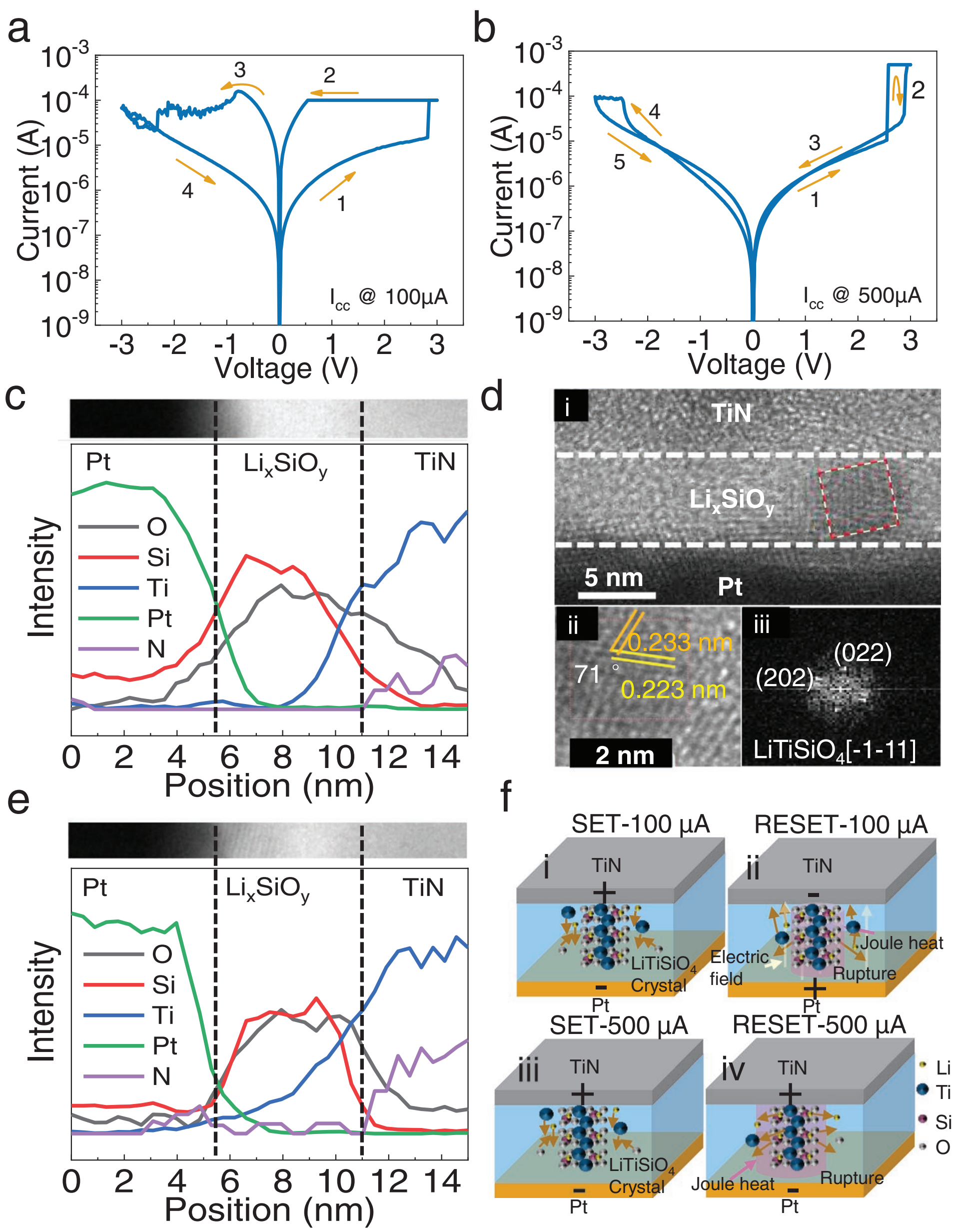

Figure 2. Artificial synaptic device characteristics. a,b) The typical *I–V* curve of device under I_{cc} of 100 µA (a) and 500 µA (b). The arrows and numbers indicate the resistive switching sequences. c) The line scan EDS spectrum of the device cross-section. d) i) the cross-sectional HRTEM image of the device, ii) zoom-in filament area, and iii) corresponding FFT image of crystalline filament. The measured angle and distance between relative crystal faces and diffraction spots suggest that the filament is crystalline lithium titanium silicate ($LiTiSiO_4$). e) The line scan EDS spectrum of the filament. The result indicates Ti and Si/O ratio profile different from the surrounding area. f) Schematic of the switching mechanism of the device under different I_{cc}.

Supporting Information). Furthermore, the threshold I_{cc} for changing the device from bipolar switching to unipolar switching was investigated by setting different I_{cc} of 100 to 400 μA circularly, as shown in Figure S4, Supporting Information. The results suggest the threshold I_{cc} would be 400 μA. After the SET operation with 100 μA I_{cc}, the line scan of energy dispersive spectrum (EDS) and element mapping analysis are used to identify the elements and structure of the cross-section, as shown in Figure 2c and Figure S5, Supporting Information, respectively. The results further identify the elemental composition and structure of the device. Furthermore, both of the line scan and mapping results indicate that the element Ti was injected into lithium silicate layer, which is similar to the results of previous work with TiN as electrode.[42] It should be noted that the Li element is hard to be observed by EDS analysis due to the ultrasoft Li K X-ray at around 55 eV.[43]

To understand the switching mechanism, the device operated at I_{cc} = 100 μA is characterized by HRTEM, as shown in Figure 2d. Figure 2d(i) shows that there is a suspicious filament region of ≈5 nm in the lithium silicate layer (box). Lattice fringes and corresponding fast Fourier transform (FFT) image of the filament are presented in Figure 2d(ii) and d(iii), respectively. The measured angle and distance between relative crystal faces is ≈71°, 0.233, and 0.223 nm, indicating that the filament is in a crystalline lithium titanium silicate phase ($LiTiSiO_4$). Because there are states density at the Fermi level of the $LiTiSiO_4$, which results in a zero band gap, the $LiTiSiO_4$ is a conductor.[44] Hence, the formed $LiTiSiO_4$ filament induces the switching from HRS to LRS of the device. Generally, there are two mechanisms for filament rupture, one is dominated by electrochemical effect, the filament rupture (RESET) and formation (SET) process is driven by opposite electric field, showing bipolar switching. The other is dominated by Joule heat effect, and the filament rupture process is independent of electric field polarity. The produced Joule heat under 100 μA compliance current is not sufficient to rupture the filament, so that the device exhibits bipolar resistive switching property. However, when the I_{cc} is 500 μA, the Joule heat would be increased located at the filament region, inducing the rupture of the filament. Thus, the device exhibits unipolar resistive switching behavior. To further understand the switching mechanism of this device, the EDS line scan of the filament area was performed, as illustrated in Figure 2e. As aforementioned, the Li element is hard to be observed by EDS analysis. However, the migration and redox of Li under the electric field have been widely investigated for battery applications due to the low ionization energy of Li.[45–46] In previous works, the migration of Li ions and oxygen vacancies in Pt/LiSiO/TiN and Pt/LiSiO/Ti was believed to induce the resistive switching.[39–41] Also, the Ti ion migration was reported to contribute to the switching process of the $TiN/Pr_{0.7}Ca_{0.3}MnO_3/Pt$ device.[47] In our work, the EDS of Ti and Si/O ratio shows obvious differences between the out-filament area and filament area (Figure 2c,e). The Ti profile of the filament area extended to the bottom electrode, and the Si/O ratio decreased, which indicates that the Ti migrated toward the bottom electrode and oxygen agglomerated towards the filament area. According to the above analysis, although without the direct observation of Li, we believe that the cooperation of Li, Ti, and O attributes the resistive switching of the device. In detail, when the I_{cc} is 100 μA (bipolar resistive switching), during the SET process, the positive voltage induces the migration of Ti and Li ions toward to bottom electrode and the positively charged ions attract surrounding suspended oxygen ions to form the filament ($LiTiSiO_4$). When negative voltage stress is imposed on the device, Ti and Li ions of filament move reversely under the electric field. The electric field cooperates with the Joule heat, which prompts the thermorphoresis/diffusion of ions, leading to the rupture of the filament.[48] When the I_{cc} is 500 μA (unipolar resistive switching), after the initial SET process, Joule heat would be dominated, leading to the rupture of the filament. The Joule heat effect has been widely reported as the unipolar switching mechanism.[49] To clearly present these two different switching behaviors, the switching schematics of the device under different I_{cc} are illustrated in Figure 2f. Initially, the device is at HRS since the lithium silicate film is in a low conductive amorphous phase. When a positive voltage is applied on the device under a low I_{cc} (100 μA), a conductive filament composed of crystalline lithium titanium silicate is formed and the device switches from HRS to LRS (Figure 2f(i)). The filament cannot be dissolved owing to insufficient joule heat. To rupture the filament, a negative voltage is required to apply on the device, which could induce the co-effect of Joule heating and electric field (Figure 2e(ii)). However, under a high I_{cc} (500 μA), the crystalline filament is formed first (Figure 2f(iii)), and then ruptured due to the considerable Joule heating effect (Figure 2f(iv)).

According to the above device characteristics, pulse stimuli with different amplitudes and widths are used to implement long-term plasticity and habituation characteristics following the dual-process theory.[38] Pulses with nanosecond-level width and relatively lower amplitude were used to implement long-term plasticity (see Figures S6 and S7, Supporting Information). What interesting is that when the positive pulse amplitude is increased to 2.3 V (width 100 ns), the conductance of the device does not increase monotonously but shows a slight decrease tendency under the repetitive stimuli, which is a weak habituation process (see Figure S8, Supporting Information). Hence, pulses with a wider width and higher amplitude were used to implement the significant habituation characteristics.

To further connect with the neuron circuit and match its parameters,[50] stimuli with different frequencies (1k, 333.3, and 200 Hz, **Figure 3**a–c) and intensities (3.2, 3.4, and 3.6 V, Figure S9a–c, Supporting Information) are used to test the pulse responses of the device. Comparing with high frequency stimuli, the device needs more stimuli to start habituation under low frequency stimuli (4, 31, and 68 pulses for 1k, 333.3, and 200 Hz, respectively). It means that the device habituates faster at higher frequency stimuli, which is consistent with the stimulus-frequency-dependence of habituation (definitions of habituation are listed in Note S1, Supporting Information). As shown in Figure S9a–c, Supporting Information, the device habituates faster to lower intensity stimuli (stimulus intensity dependence of habituation). To eliminate the randomness in the frequency and intensity test, the pulse number for device habituating under different stimulus trains is counted (Figure 3d and Figure S9d, Supporting Information). Each stimulus train is imposed on the device for ten times. The statistical results further confirm the stimulus intensity/frequency dependence

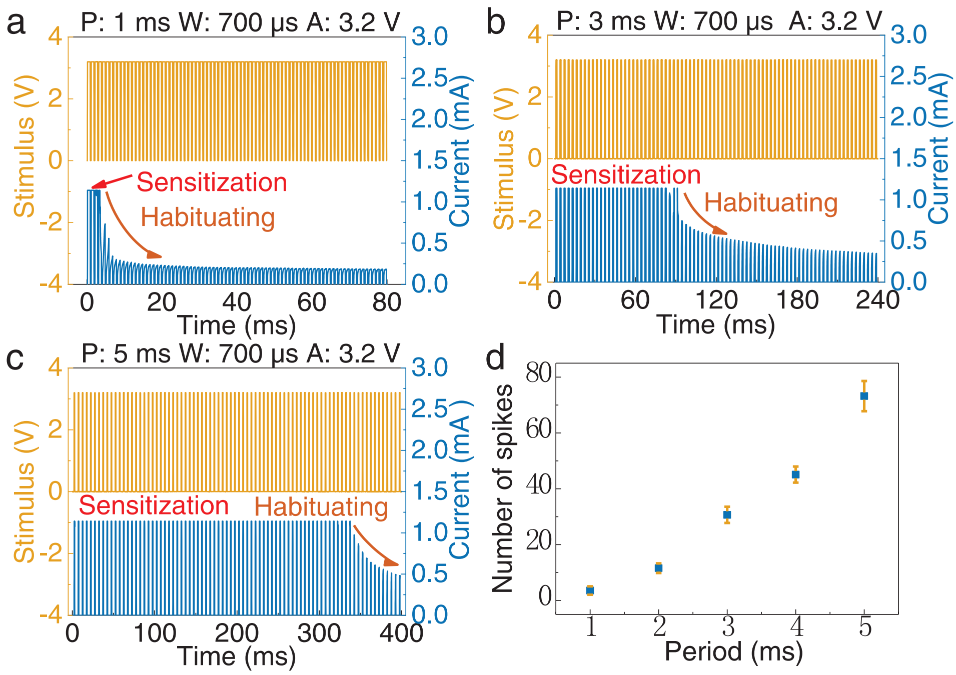

Figure 3. Habituation of the synaptic device at ms-level stimulus train. Stimulus trains with identical amplitude (A: 3.2 V), width (W: 700 µs), period (P: 1 ms) and different frequencies are imposed on the device. The real-time stimulus train and the corresponding response currents under frequencies of 1 kHz (a); 333.3 Hz (b); and 200 Hz (c). The response current of stimulus trains at high frequency retains a shorter time before degradation than that at low frequency. This means that the synapse habituates faster to high frequency stimuli. d) Statistical data of the pulse number before degradation as a function of stimulus frequency, each data point is collected from ten trials.

of the device. Additionally, repetitive stimulus is essential for the device to habituate, non-repetitive stimulus cannot induce the habituation behavior (see Figure S10, Supporting Information). It should be noted that the sensitization process is too fast to be presented under the millisecond-level stimuli. Thus, the stimulus pulse width is reduced to microsecond-level for clearly emulating the dual-process of habituation (see Figures S11 and S12, Supporting Information). The results show that the device could successfully emulate the habituation characteristics, such as stimulus intensity/frequency dependence, potentiation of habituation, dishabituation, habituation of dishabituation and spontaneous recovery, following the dual-process theory. The definition of these characteristics is described in Note S1, Supporting Information. Based on the above results, it is concluded that the characteristics of habituation are successfully demonstrated in the TiN/Li_xSiO_y/Pt synaptic device.

To demonstrate that the habituation process can work in the SNS, the fully memristor-based SNS (schematic is shown in **Figure 4**a) is constructed by connecting the habituation synaptic device with an Ag-based LIF neuron that has been reported in our previous work.[50] The SNS works as follows: first, input pulses, representing the electrical signal transformed by the sensory neuron, are repetitively applied on the synaptic device; second, the repetitive pulses induce habituation behaviors of the synaptic device, namely the decrease of the current or conductance; finally, the voltage distributed on the relay neuron decreases owing to habituation of the synaptic device, resulting in a decreased neuron spiking frequency. The neuron circuit (Figure S13a, Supporting Information) consists of a capacitor (100 nF), an Ag-based threshold switching memristor (TSM), and an output resistor (10 kΩ). As shown in Figure S13b, Supporting Information, the TSM device switches from an HRS to an LRS when the applied voltage reaches a threshold value at ≈1 V. Afterwards, the device will spontaneously switch back to the HRS when the applied voltage is lower than a hold voltage (≈0.11 V). The Ag-based TSM shows typical threshold switching behavior, which is critical to the spiking behavior of the Ag-based LIF neuron. Figure 4b shows the spiking behavior of the neuron under a series of stimulus trains (1.8 V, 700 µs). Under input pulses, the charge and discharge of the capacitor result in the evolution of the voltage between two electrodes of the TSM device. The voltage evolution would induce the resistive switching of the TSM device between the HRS and LRS, dominating the LIF process of the neuron circuit, as shown in Figure S13a, Supporting Information. Furthermore, the output characteristics of the neuron circuit under different input resistance were investigated and shown in Figure S13c, Supporting Information. The output spiking frequency of this neuron decreases with increasing the input resistance that corresponds to the pre-synapse weight of the neural network.

For the demonstration of the output spiking frequency of the neuron with different synaptic weights, we further replace the input resistance with a synaptic device to construct the SNS. As shown in Figure 4c, a stimulus train (4.0 V, 700 µs) is performed as the input to the SNS and the response of the SNS is characterized. Evident habituation of the spiking frequency of the SNS is observed. In the time window from 0 to ≈0.1 s, the spiking frequency is very high, corresponding to the biological

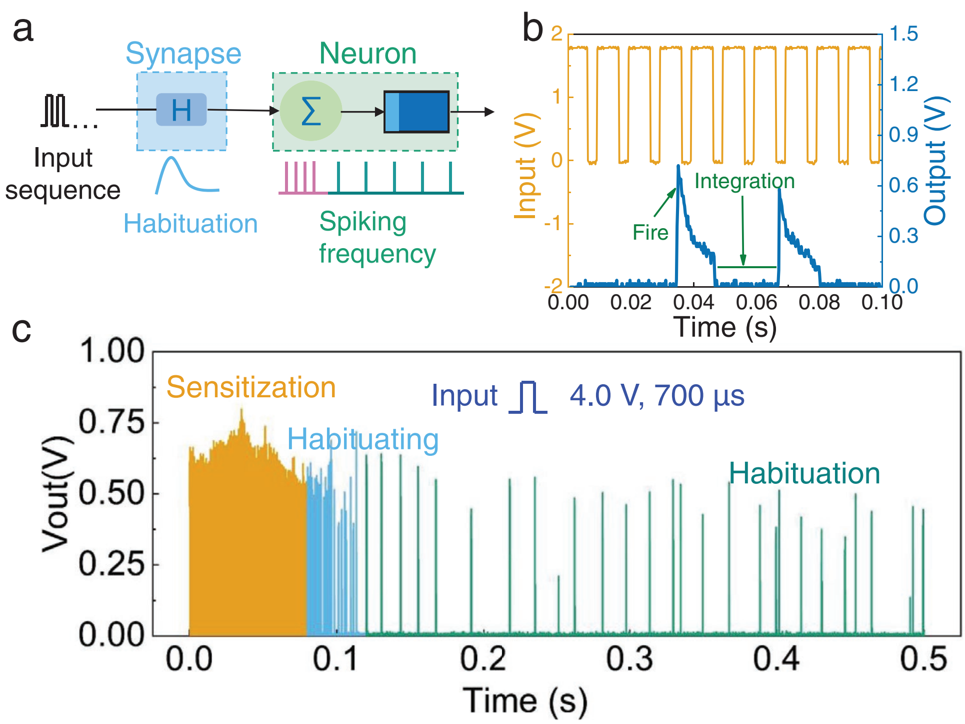

Figure 4. SNS with habituation emulated by memristive devices. a) Schematic of the SNS. b) The spiking behavior of the relay neuron under continuous input. c) The output spikes of the fully memristor-based SNS. In the beginning, the synapse is set to the LRS state, then the output of the system is at a high frequency (≈1 kHz). Afterward, the synapse is depressed to the HRS and the output of the system is at a low frequency (≈80 Hz).

sensitization process where a new stimulus is applied on the organism and results in the progressive amplification of a response.[51] However, from 0.1 to 0.5 s, the repetition of the stimulus gives rise to a habituation process, and the response (spiking frequency) decreases dramatically. The results demonstrate that the habituation is successfully achieved in the memristor-based SNS.

It should be noted that the sensory neuron could also be implemented basing on Mott memristor to construct a fully memristor based SNS (Figure S14, Supporting Information). The sensory neuron part contains a Mott memristor, a sensor (infrared distance sensor), a capacitor, a resistor, and a comparator, as shown in Figure S14a (Supporting Information). Here, the sensor was used to detect the distance information of objects. The distance measuring characteristics of the sensor is shown in Figure S14b (Supporting Information). The output of the sensor is the constant voltage when the distance between the object (white paper) and the sensor is certain. With decreasing the distance, the output voltage of the sensor is increasing. Here, a Mott memristor-based convertor was used to convert the constant voltage into spikes, which has been demonstrated in our previous work.[52] The typical I–V curve of the Mott memristor is shown in Figure S14c (Supporting Information). When the output of the sensor reaches the threshold of the Mott converter, the converter generates a serial spike (zigzag), as presented in Figure S14d (red; Supporting Information). For further convert the output (Node 1) of the Mott converter to square waves that serve as the input of the habituation synapse, a comparator was connected with the Mott converter with a reference voltage (1.5 V). When the output of Mott converter exceeds the reference voltage, the output of the comparator would be V_{cc} (4 V), whereas the output would be 0 V. The output of the comparator is illustrated in Figure S14d (blue; Supporting Information). It should be noted that the duty factor of the V_{out} could be flexible by adjusting the V_{ref}. Furthermore, the frequency of the V_{out} could be flexible by adjusting the capacitor, the input resistor of the Mott converter, and the distance between the object and sensor.[52] The results indicate that the sensory neuron function could also be implemented by combining sensors and memristor devices.

To demonstrate the potential application of the SNS, an unsupervised HSNN is proposed (Note S1, Supporting Information). In the network, a habituation learning rule extracted from the experimental data is used to accomplish the obstacle avoidance function of robots. The architecture of the HSNN for robot navigation is shown in **Figure 5**a. It consists of nine sensory neurons as the inputs, one relay neuron connected with sensory neurons through habituation synapses for information processing and one motor neuron to control the movement of the robot. As mentioned above, sensory neurons can be seen as transducers and no information processing occurs. Thus, sensory neurons are represented as inputs to the HSNN. The input is the electrical signal (3.2 V, 700 μs, and 333.3 kHz, the same as in Figure 3b) representing an obstacle in front of the robot. Moreover, every three sensory neurons are grouped as a receptive field,[53] and it is assumed that at any time, only one receptive field, which is nearest to the obstacle, is active. Collision is defined as the detection of the obstacle by three sensory neurons at the same time. Thus, one collision corresponds to the activation of one receptive field. The weight change of

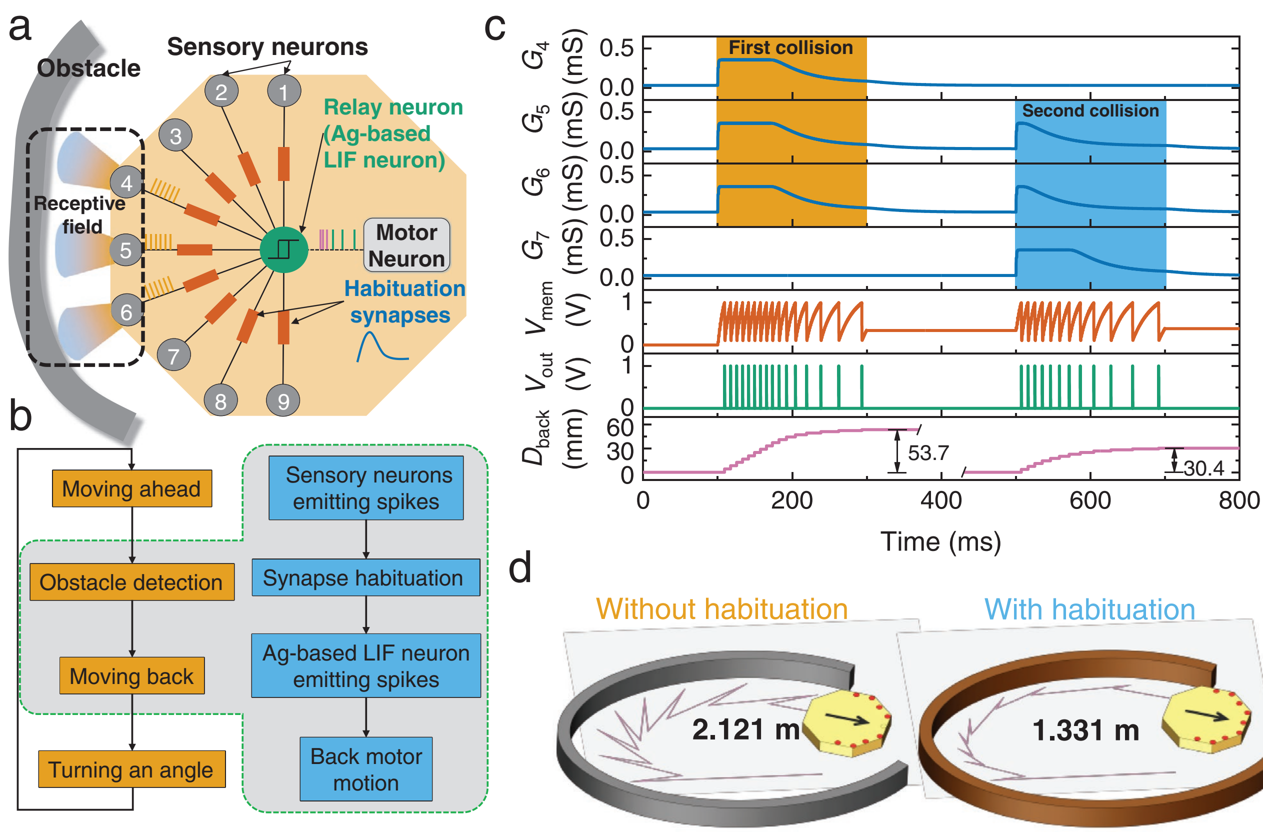

Figure 5. HSNN for obstacle avoidance in robot navigation. a) Schematic of HSNN. Nine sensory neurons are used as the inputs, which are connected with a relay neuron (Ag-based LIF neuron) through synapses with habituation characteristics. Spiking of the relay neuron is transmitted to the motor neuron and its frequency is used to control the backward movement of the robot. b) Flow diagram of robot obstacle avoidance with habituation mechanism. c) The evolution of relative parameters of SNS under two collision situations. G_n is the conductance of habituation synapse n. V_{mem} is the membrane potential of Ag-based LIF neuron. V_{out} is the emitting spikes of the LIF neuron. D_{back} is the backward distance of the robot. d) Illustration of the motion trajectory of the robot without and with habituation when avoiding obstacles. The robot with habituation has a smoother trajectory and can avoid continuous obstacles more effectively. The distance traveled by the robot with habituation is 1.331 m, which is about 37.2% reduction of the distance without habituation (2.121 m).

the synapses linking the sensory neuron and the relay neuron follows the habituation rule extracted from the synaptic device (see Note S3, Supporting Information). The relay neuron is a LIF neuron. It emits a spike once its membrane potential (the potential at node 1 in Figure 1c) surpasses a threshold value of 1 V and stops spiking when the membrane potential drops to 0.11 V. The threshold values are extracted from the experimental data (see Figure S13b, Supporting Information). The spiking outputs of the relay neuron are transmitted to the motor neuron. The motor neuron transforms the spiking signals to the backward movement of the robot and the backward distance D_{back} is proportional to the spiking frequency f with a relation $D_{back} = \alpha \times f$. Where α is the proportional coefficient and the value is about 1/30. For simplification, a constant forward movement of the robot is assumed. After each collision with the obstacle, the robot moves back and turns a small angle (25°) to avoid the obstacle. The detailed obstacle avoidance process of the robot is illustrated in Note S4, Supporting Information.

The flow chart of the simulation in one collision to the obstacle is shown in Figure 5b and Figure S15, Supporting Information: first, the robot moves forward at a constant speed; once an obstacle is detected, the sensory neurons emit spikes to the relay neuron via the habituation synapses; then the relay neuron fires to make the robot move back and turn an constant angle to avoid the obstacle. Then the evolution of the network parameters with habituation characteristics is shown in Figure 5c. In the first collision, the receptive field composed of sensory neurons 4, 5, and 6 is active. The emitted spikes of the sensory neurons induce synaptic weight change following the habituation rule, as shown in the top three curves G_4, G_5, and G_6 of Figure 5c. The weight change modulates the current flowing through the synaptic device, influences the charging speed of the capacitor in the relay neuron, then the membrane potential (V_{mem}) of the relay neuron is regulated. Finally, the output spiking numbers (frequency) of the relay neuron (V_{out}) is tuned according to the V_{mem}. The output spiking number controls the backward movement of the robot. As shown in Figure 5c, the habituation behavior of the output spiking numbers is clearly observed in one collision. After turning a small angel, a second collision is detected by the adjacent receptive field (sensory neurons 5, 6, and 7). Since overlapped neurons 5 and 6 are already habituated in the first collision, they habituate faster in the second collision according to the potential of habituation characteristic (see Note S2 and Figure S11c, Supporting Information). Therefore, the V_{mem} and V_{out} in the second collision is suppressed in comparison with the first one, leading to a short backward distance. As shown in the bottom panel of Figure 5c, the backward distance in the first collision is 53.7 mm, but in the second collision, it is reduced to 30.4 mm. It should be noted that the number of overlapped sensory neurons in the receptive field between two adjacent collisions could be 0, 1, 2, 3. The evolution of the network parameters with the situations of 0, 1, 3 overlapped sensory neurons are shown in Figure S16, Supporting Information. For comparison, we performed the simulation without synaptic habituation, and the evolution of the network parameters are shown in Figure S17, Supporting Information. A fixed distance for each backward action for all cases is observed. Figure 5d shows the trajectories of the robot without and with habituation characteristics when avoiding obstacles. The robot with habituation has a smoother trajectory and shorter traveled distance of 1.331 m than that of 2.121 m for the situation without habituation (37.2% reduction). It can avoid continuous obstacles more effectively. The results show that the habituation learning mechanism enables the robot to ignore a repetitive stimulus that could slow it down in its path course. Thus, the robot can achieve better performance in the scenario of obstacle avoidance.

In this work, a fully memristor-based SNS with habituation characteristics is demonstrated for the first time. It consists of a synapse (TiN/Li_xSiO_y/Pt) and an Ag-based neuron. Besides conventional synaptic plasticity, the synaptic device shows habituation behaviors obeying its biological definition. By connecting the synaptic device to the LIF neuron, the spiking frequency of the SNS is habituated to the repetitive input stimulus. Finally, a HSNN based on the experimental results of the SNS is built and its function is successfully demonstrated in a scenario of obstacle avoidance in robot navigation. With the HSNN, the robot can filter out repetitive stimulus that slows it down in navigation. The results show that a direct emulation of the biological habituation rule by memristors is critical for SNS to adapt to a dynamic environment. Furthermore, developing the habituation neural network is favorable to novelty detection, recency encoding, temporal signal processing, etc.

Experimental Section

Fabrication of Samples: The fabrication processes of synaptic devices include: 1) After the first lithography process, the bottom electrode Pt/Ti (35/5 nm) was deposited by e-beam evaporation and released by the first lift-off process. 2) After the second lithography process, the dielectric layer Li_xSiO_y (5 nm) was deposited by magnetron sputtering and released by the second lift-off process. 3) After the third lithography process, the top electrode TiN (40 nm) was deposited by ion beam sputtering and released by the third lift-off process.

The fabrication processes of Ag-based threshold devices include: 1) The deposition process of bottom electrode Au/Ti (35/5 nm) was similar to the process of synaptic device bottom electrodes. 2) After the lithography process, the dielectric layer SiO_2:Ag (10 nm) was grown by co-sputtering and released by the second lift-off process. 3) The deposition process of top electrode Au/Ag (10/30 nm) was similar to the process of synaptic device bottom electrode.

The fabrication process of Mott memristors includes: 1) The bottom electrode TiN (40 nm) was deposited by ion beam sputtering. Then the first lithography process was used to pattern the bottom electrode followed by wet etching (H_2O_2, 30 min) process. 2) After the second lithography process, the dielectric layer NbO_x (50 nm) was deposited by magnetron sputtering and released by lift-off process. 3) The deposition process of top electrode TiN (40 nm) was similar to the process of synaptic device top electrodes.

The effective device area of all the fabricated devices in this work are 5×5 μm^2.

Characterization: The XRD and XPS results were obtained on Rigaku D/max-2500 and ESCALAB 250Xi, respectively. The DC-mode and pulse mode characteristics of the devices were measured with an Agilent B1500A Semiconductor Characterization System. The electrical characteristics of the memristive SNS was measured using the AFG3102 Arbitrary Function Generator as a signal source and MSO3032 Mixed Signal Oscilloscope to collect output signals.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

habituation, memristors, robot navigation, sensory nervous system

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